

**Mõõtereled ja kaitseaparatuur. Osa 26:
Elektromagnetilise ühilduvuse nõuded**

**Measuring relays and protection equipment - Part 26:
Electromagnetic compatibility requirements**

EESTI STANDARDI EESSÕNA

NATIONAL FOREWORD

See Eesti standard EVS-EN 60255-26:2013 sisaldab Euroopa standardi EN 60255-26:2013 ingliskeelset teksti.	This Estonian standard EVS-EN 60255-26:2013 consists of the English text of the European standard EN 60255-26:2013.
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**Measuring relays and protection equipment -
Part 26: Electromagnetic compatibility requirements
(IEC 60255-26:2013)**

Relais de mesure et dispositifs de
protection -
Partie 26: Exigences de comptibilité
électromagnétiques
(CEI 60255-26:2013)

Messrelais und Schutzeinrichtungen -
Teil 26: Anforderungen an die
elektromagnetische Verträglichkeit
(IEC 60255-26:2013)

This European Standard was approved by CENELEC on 2013-06-28. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

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European Committee for Electrotechnical Standardization
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Europäisches Komitee für Elektrotechnische Normung

CEN-CENELEC Management Centre: Avenue Marnix 17, B - 1000 Brussels

Foreword

The text of document 95/309/FDIS, future edition 3 of IEC 60255-26, prepared by IEC/TC 95 "Measuring relays and protection equipment" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN 60255-26:2013.

The following dates are fixed:

- latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2014-03-28
- latest date by which the national standards conflicting with the document have to be withdrawn (dow) 2016-06-28

This document supersedes EN 60255-11:2010 (PART), EN 60255-22-1:2008, EN 60255-22-2:2008, EN 60255-22-3:2008, EN 60255-22-4:2008, EN 60255-22-5:2011, EN 60255-22-6:2001, EN 60255-22-7:2003, EN 60255-25:2000, EN 60255-26:2009

EN 60255-26:2013 includes the following significant technical changes with respect to EN 60255-26:2009:

- a) definition of test specifications, test procedures and acceptance criteria per phenomena and port under test in one document;
- b) extension of radiated emission measurement for frequencies above 1 GHz;
- c) limitation of radiated emission measurement at 3 m distance for small equipment only;
- d) addition of zone A and zone B test level on surge test;
- e) extension of tests on the auxiliary power supply port by a.c. and d.c. voltage dips, a.c. component in d.c. (ripple) and gradual shut-down / start-up;
- f) harmonization of acceptance criteria for immunity tests.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CENELEC [and/or CEN] shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For the relationship with EU Directive(s) see informative Annex ZZ, which is an integral part of this document.

Endorsement notice

The text of the International Standard IEC 60255-26:2013 was approved by CENELEC as a European Standard without any modification.

In the official version, for Bibliography, the following note has to be added for the standard indicated:

IEC 61000-4

NOTE Harmonized in EN 61000-4 series (not modified)

Annex ZA
(normative)
**Normative references to international publications
with their corresponding European publications**

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE When an international publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD</u>	<u>Year</u>
IEC 60255-1	2009	Measuring relays and protection equipment - EN 60255-1 Part 1: Common requirements	EN 60255-1	2010
IEC 61000-4-2	2008	Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test	EN 61000-4-2	2009
IEC 61000-4-3 + A1 + A2	2006 2007 2010	Electromagnetic compatibility (EMC) - Part 4-3: Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test	EN 61000-4-3 + A1 + A2	2006 2008 2010
IEC 61000-4-4	2012	Electromagnetic compatibility (EMC) - Part 4-4: Testing and measurement techniques - Electrical fast transient/burst immunity test	EN 61000-4-4	2012
IEC 61000-4-5 + corr. October	2005 2009	Electromagnetic compatibility (EMC) - Part 4-5: Testing and measurement techniques - Surge immunity test	EN 61000-4-5	2006
IEC 61000-4-6	2008	Electromagnetic compatibility (EMC) - Part 4-6: Testing and measurement techniques - Immunity to conducted disturbances, induced by radio-frequency fields	EN 61000-4-6	2009
IEC 61000-4-8	2009	Electromagnetic compatibility (EMC) - Part 4-8: Testing and measurement techniques - Power frequency magnetic field immunity test	EN 61000-4-8	2010
IEC 61000-4-11	2004	Electromagnetic compatibility (EMC) - Part 4-11: Testing and measurement techniques - Voltage dips, short interruptions and voltage variations immunity tests	EN 61000-4-11	2004
IEC 61000-4-16 + A2	1998 2009	Electromagnetic compatibility (EMC) - Part 4-16: Testing and measurement techniques - Test for immunity to conducted, common mode disturbances in the frequency range 0 Hz to 150 kHz	EN 61000-4-16 + A2	1998 2011
IEC 61000-4-17 + A1 + A2	1999 2001 2008	Electromagnetic compatibility (EMC) - Part 4-17: Testing and measurement techniques - Ripple on d.c. input power port immunity test	EN 61000-4-17 + A1 + A2	1999 2004 2009

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD</u>	<u>Year</u>
IEC 61000-4-18 + A1	2006 2010	Electromagnetic compatibility (EMC) - Part 4-18: Testing and measurement techniques - Damped oscillatory wave immunity test	EN 61000-4-18 + corr. September + A1	2007 2007 2010
IEC 61000-4-29	2000	Electromagnetic compatibility (EMC) - Part 4-29: Testing and measurement techniques - Voltage dips, short interruptions and voltage variations on d.c. input power port immunity tests	EN 61000-4-29	2000
CISPR 11 (mod) + A1	2009 2010	Industrial, scientific and medical equipment - Radio-frequency disturbance characteristics - Limits and methods of measurement	EN 55011 + A1	2009 2010
CISPR 22 (mod)	2008	Information technology equipment - Radio disturbance characteristics - Limits and methods of measurement	EN 55022 + AC:2011	2010 2011

Annex ZZ
(informative)
Coverage of Essential Requirements of EC Directives

This European Standard has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association and within its scope the standard covers all relevant essential requirements as given in Article 1(a) of Annex I of EC Directive 2004/108/EC.

Compliance with this standard provides one means of conformity with the specified essential requirements of the Directive concerned.

WARNING: Other requirements and other EC Directives may be applicable to the products falling within the scope of this standard.

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INTRODUCTION

This part of the IEC 60255 series specifies all of the requirements for electromagnetic compatibility in a single document.

As such, it is considered as an overview document for measuring relays and protection equipment. The detailed test procedures are given in other referenced standards.

This part of IEC 60255 does not include the reversal of d.c. power supply polarity test which had been provided in IEC 60255-11, because this is a safety test. This test will be covered by future IEC 60255-27.

MEASURING RELAYS AND PROTECTION EQUIPMENT –

Part 26: Electromagnetic compatibility requirements

1 Scope

1.1 General

This part of the IEC 60255 series is applicable to measuring relays and protection equipment, taking into account combinations of devices to form schemes for power system protection including the control, monitoring, communication and process interface equipment used with those systems.

This standard specifies the requirements for electromagnetic compatibility for measuring relays and protection equipment.

Tests specified in this standard are not required for equipment not incorporating electronic circuits, for example electromechanical relays.

The requirements specified in this standard are applicable to measuring relays and protection equipment in a new condition and all tests specified are type tests only.

1.2 Emission

The object of this standard is to specify limits and test methods, for measuring relays and protection equipment in relation to electromagnetic emissions which may cause interference in other equipment.

These emission limits represent electromagnetic compatibility requirements and have been selected to ensure that the disturbances generated by measuring relays and protection equipment, operated normally in substations and power plants, do not exceed a specified level which could prevent other equipment from operating as intended.

Test requirements are specified for the enclosure and auxiliary power supply ports.

1.3 Immunity

This standard is to specify the immunity test requirements for measuring relays and protection equipment in relation to continuous and transient, conducted and radiated disturbances, including electrostatic discharges.

These test requirements represent the electromagnetic compatibility immunity requirements and have been selected so as to ensure an adequate level of immunity for measuring relays and protection equipment, operated normally in substations and power plants.

NOTE 1 Safety considerations are not covered in this standard.

NOTE 2 In special cases, situations will arise where the levels of disturbance could exceed the levels specified in this standard, for example where a hand-held transmitter or a mobile telephone is used in close proximity to measuring relays and protection equipment. In these instances, special precautions and procedures could have to be employed.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60255-1:2009, *Measuring relays and protection equipment – Part 1: Common requirements*

IEC 61000-4-2:2008, *Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques – Electrostatic discharge immunity test*

IEC 61000-4-3:2006, *Electromagnetic compatibility (EMC) – Part 4-3: Testing and measurement techniques – Radiated, radio-frequency, electromagnetic field immunity test*
Amendment 1:2007
Amendment 2:2010

IEC 61000-4-4:2012, *Electromagnetic compatibility (EMC) – Part 4-4: Testing and measurement techniques – Electrical fast transient/burst immunity test*

IEC 61000-4-5:2005, *Electromagnetic compatibility (EMC) – Part 4-5: Testing and measurement techniques – Surge immunity test*

IEC 61000-4-6:2008, *Electromagnetic compatibility (EMC) – Part 4-6: Testing and measurement techniques – Immunity to conducted disturbances, induced by radio-frequency fields*

IEC 61000-4-8:2009, *Electromagnetic compatibility (EMC) – Part 4-8: Testing and measurement techniques – Power frequency magnetic field immunity test*

IEC 61000-4-11:2004, *Electromagnetic compatibility (EMC) – Part 4-11: Testing and measurement techniques – Voltage dips, short interruptions and voltage variations immunity tests*

IEC 61000-4-16:1998, *Electromagnetic compatibility (EMC) – Part 4-16: Testing and measurement techniques – Test for immunity to conducted, common mode disturbances in the frequency range 0 Hz to 150 kHz*
Amendment 2:2009

IEC 61000-4-17:1999, *Electromagnetic compatibility (EMC) – Part 4-17: Testing and measurement techniques – Ripple on d.c. input power port immunity test*
Amendment 1:2001
Amendment 2:2008

IEC 61000-4-18:2006, *Electromagnetic compatibility (EMC) – Part 4-18: Testing and measurement techniques – Damped oscillatory wave immunity test*
Amendment 1:2010

IEC 61000-4-29:2000, *Electromagnetic compatibility (EMC) – Part 4-29: Testing and measurement techniques – Voltage dips, short interruptions and voltage variations on d.c. input power port immunity tests*

CISPR 11:2009, *Industrial, scientific and medical equipment – Radio-frequency disturbance characteristics – Limits and methods of measurement*
Amendment 1:2010

CISPR 22:2008, *Information technology equipment – Radio disturbance characteristics – Limits and methods of measurement*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1 EUT

Equipment Under Test

equipment which may be either a measuring relay or protection equipment

3.2 small equipment

equipment, either positioned on a table top or standing on the floor which, including its cables fits in a cylindrical test volume of 1,2 m in diameter and 1,5 m above the ground plane.

3.3 port

particular interface of the specified EUT with the external electromagnetic environment

SEE: Figure 1

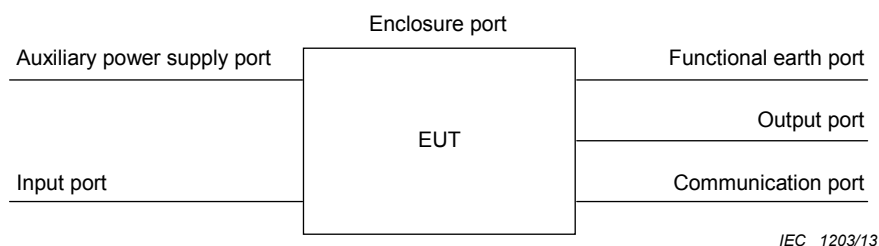


Figure 1 – Ports for measuring relays and protection equipment

3.4

auxiliary power supply port

a.c. or d.c. auxiliary energizing input of the EUT

3.5

communication port

interface with a communication and/or control system permanently connected to the EUT

3.6

enclosure port

physical boundary of the EUT through which electromagnetic fields may radiate or impinge

3.7

functional earth port

port on the EUT which is connected to earth for purposes other than electrical safety

3.8

input port

port through which the EUT is energized or controlled in order to perform its function(s)

EXAMPLE Current and voltage transformer, binary input, etc.

3.9

output port

port through which the EUT produces predetermined changes

EXAMPLE Contacts, optocouplers, analogue output, etc.

3.10

coupling/decoupling network

CDN

electrical circuit for the purpose of transferring energy from one circuit to another/
electrical circuit for the purpose of preventing test voltages applied to the EUT from affecting other devices, equipment, or systems which are not under test

3.11

common mode

CM

mode between each active conductor and a specified reference, usually earth or ground reference plane

3.12

differential mode

DM

mode between any two of a specified set of active conductors

4 Definition of environmental levels

4.1 General

The environmental levels shall be selected in accordance with the most realistic installation and environmental conditions in which the EUT is expected to operate.

Based on common installation practices, the recommended selection of test levels is the following:

4.2 Zone A, severe electrical environment

The installation is characterized by the following attributes:

- no suppression of electrical fast transients/bursts in the power supply and control and power circuits which are switched by relays and contactors;
- no separation of the industrial circuits from other circuits associated with environments of higher severity levels;
- no separation between power supply, control, signal and communication cables;
- use of common multi-core cables for control and signal lines.

The outdoor area of industrial process equipment, where no specific installation practice has been adopted, of power stations, open-air HV substation switchyards and gas insulated switchgear may be representative of this environment.

4.3 Zone B, typical electrical environment

The installation is characterized by the following attributes:

- no suppression of electrical fast transients/bursts in the power supply and control circuits which are switched by relays (no contactors);
- poor separation of the industrial circuits from other circuits associated with environments of higher severity levels;

- dedicated cables for power supply, control, signal and communication lines;
- poor separation between power supply, control, signal and communication cables;
- availability of earthing system represented by conductive pipes, ground conductors in the cable trays (connected to the protective earth system) and by a ground mesh.

The area of industrial process equipment, the power plants and the relay room of open-air HV substations may be representative of this environment.

5 Emission

5.1 Emission enclosure

Table 1 – Emission tests – Enclosure port

Item	Environmental phenomena	Frequency range	Limits	Basic standard	Test procedure
1.1	Radiated emission (below 1 GHz)^{ab}	30 MHz to 230 MHz	40 dB(μV/m) quasi peak at 10 m 50 dB(μV/m) quasi peak at 3 m	CISPR 11	See 7.1.2
		230 MHz to 1 000 MHz	47 dB(μV/m) quasi peak at 10 m 57 dB(μV/m) quasi peak at 3 m		
1.2	Radiated emission (above 1 GHz)	1 GHz to 3 GHz	56 dB(μV/m) average 76 dB(μV/m) peak at 3 m	CISPR 22	See 7.1.2
		3 GHz to 6 GHz	60 dB(μV/m) average 80 dB(μV/m) peak at 3 m		
^a Measuring relays and protection equipment are apparatus which satisfy the class A limits. Limits can be measured at a nominal distance of 3 m, 10 m or 30 m. A measuring distance less than 10 m is allowed only for equipment which complies with the definition given in 3.2. In case of measurements at a separation distance of 30 m an inverse proportionality factor of 20 dB per decade shall be used to normalize the measured data to the specified distance for determining compliance.					
^b The limits specified for the 3 m separation distance apply only to small equipment meeting the size criterion defined in 3.2.					

Conditional testing procedure

- The highest internal source of an EUT is defined as the highest frequency generated or used within the EUT or on which the EUT operates or tunes.
- If the highest frequency of the internal sources of the EUT is less than 108 MHz, the measurement shall only be made up to 1 GHz.
- If the highest frequency of the internal sources of the EUT is between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.
- If the highest frequency of the internal sources of the EUT is between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.
- If the highest frequency of the internal sources of the EUT is above 1 GHz, the measurement shall be made up to 5 times the highest frequency or 6 GHz, whichever is less.

5.2 Emission auxiliary power supply port

See Table 2.

Table 2 – Emission tests – Auxiliary power supply port

Item	Environmental phenomena	Frequency range	Limits	Basic standard	Test procedure
2.1	Conducted emission	0,15 MHz to 0,50 MHz	79 dB(μV) quasi peak 66 dB(μV) average	CISPR 22	See 7.1.3
		0,5 MHz to 30 MHz	73 dB(μV) quasi peak 60 dB(μV) average		

6 Immunity

6.1 Immunity enclosure

Table 3 – Immunity tests – Enclosure port

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
3.1	Radiated radiofrequency electromagnetic field					
	Frequency sweep Test frequency range Test field strength (prior to modulation) Amplitude modulated Sweep rate	80 to 1 000 1 400 to 2 700 10 80 ≤ 1	MHz MHz V/m (r.m.s.) % AM (1 kHz) % (of preceding frequency value)			
	Spot frequencies Test spot frequencies Test field strength (prior to modulation) Amplitude modulated Duty cycle	80 ± 0,5 % 160 ± 0,5 % 380 ± 0,5 % 450 ± 0,5 % 900 ± 5 1 850 ± 5 2 150 ± 5 10 80 100	MHz MHz MHz MHz MHz MHz MHz V/m (r.m.s.) % AM (1 kHz) %	IEC 61000-4-3	7.2.4	A
3.2	Electrostatic discharge					
	Contact	2 4 6	kV (charge voltage)	IEC 61000-4-2	See 7.2.3	B
	Air	2 4 8	kV (charge voltage)			
3.3	Power frequency magnetic field					
	Continuous	30	A/m	IEC 61000-4-8	See 7.2.10	A
	1 s to 3 s	300	A/m			B

6.2 Immunity auxiliary power supply port

Table 4 – Immunity tests – Auxiliary power supply port

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
4.1	Conducted disturbance induced by radio-frequency fields					
	Frequency sweep					
	Test frequency range	0,15 to 80	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Sweep rate	≤ 1	% (of preceding frequency value)	IEC 61000-4-6	See 7.2.8	A
	Spot frequencies					
	Test spot frequencies	27 $\pm$ 0,5 % 68 $\pm$ 0,5 %	MHz MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
4.2	Fast transients					
	Rise time t_r / duration time t_d	5 / 50	ns			
	Test level			IEC 61000-4-4	See 7.2.5	B
	Zone A	4	kV peak voltage			
	Zone B	2	kV peak voltage			
4.3	Slow damped oscillatory wave					
	Voltage oscillation frequency	1	MHz			
	Test level					
	Differential mode	1	kV peak voltage	IEC 61000-4-18	See 7.2.6	B
	Common mode	2,5	kV peak voltage			
	Voltage rise time	75	ns			
	Repetition frequency	400	Hz			
	Output impedance	200	Ω			

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
4.4	Surge Front time / time to half value Source impedance Test level Line-to-line Zone A Zone B Coupling resistor Coupling capacitor Line-to-earth Zone A Zone B Coupling resistor Coupling capacitor	1,2 / 50 (8 / 20) 2 0,5 1 2 0,5 1 0 18 0,5 1 2 4 0,5 1 2 10 9	µs voltage (current) Ω kV kV Ω µF kV kV Ω µF	IEC 61000-4-5	See 7.2.7	B
4.5	A.C. and d.c. voltage dips Test level Duration time a.c. d.c. Test level Duration time a.c. d.c. Test level Duration time a.c. d.c.	 0 0,5 to 25 ^a 10 to 1 000 ^b 40 10/12 ^d 200 70 25/30 ^d 500	 % residual voltage cycles ms % residual voltage cycles ms % residual voltage cycles ms	 IEC 61000-4-11 IEC 61000-4-29 IEC 61000-4-11 IEC 61000-4-29 IEC 61000-4-11 IEC 61000-4-29	 See 7.2.11 See 7.2.11 See 7.2.11	 A C C
4.6	A.C. and d.c. voltage interruptions Test level Duration time a.c. d.c.	 0 250/300 5	 % residual voltage cycles s	 IEC 61000-4-11 IEC 61000-4-29	See 7.2.11	C
4.7	A.C. component in d.c. (ripple) Test level Test frequency	 15 % of rated d.c. value 100/120 ^c	 V Hz, sinusoidal waveform.	IEC 61000-4-17	See 7.2.12	A

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
4.8	Gradual shut-down/start-up (for d.c. power supply)					
	Shut-down ramp	60	s	---	See 7.2.13	C
	Power off	5	min			
	Start-up ramp	60	s			
a Manufacturer shall declare the duration among the following values: 0,5 cycle, 1 cycle, 2,5 cycles, 5 cycles, 10 cycles or 25 cycles. b Manufacturer shall declare the duration among the following values: 10 ms, 20 ms, 30 ms, 50 ms, 100 ms, 200 ms, 300 ms, 500 ms or 1 000 ms. c Test shall be done at a frequency of twice the specified power system frequency(s). d “10/12 cycles” means “10 cycles for 50 Hz test” and “12 cycles for 60 Hz test”						

6.3 Immunity communication port

Table 5 – Immunity tests – Communication port

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
5.1	Conducted disturbance induced by radio-frequency fields					
	Frequency sweep			IEC 61000-4-6	See 7.2.8	A
	Test frequency range	0,15 to 80	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Sweep rate	≤ 1	% (of preceding frequency value)			
	Spot frequencies			IEC 61000-4-6	See 7.2.8	A
	Test spot frequencies	27 $\pm$ 0,5 %	MHz			
		68 $\pm$ 0,5 %	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Duty cycle	100	%			
5.2	Fast transients					
	Rise time t_r / Duration time t_d	5 / 50	ns	IEC 61000-4-4	See 7.2.5	B
	Test level					
	Zone A	2	kV peak voltage			
	Zone B	1	kV peak voltage			
	Repetition frequency	5	kHz			

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
5.3	Slow damped oscillatory wave Voltage oscillation frequency Test level Differential mode Common mode Voltage rise time Repetition frequency Output impedance	 1 0 1 75 400 200	 MHz kV peak voltage kV peak voltage ns Hz Ω	 IEC 61000-4-18 	 See 7.2.6 	 B
5.4	Surge Front time / time to half value Source impedance Test level Line-to-earth Zone A Zone B Coupling resistor ^a Coupling capacitor ^a	 1,2 / 50 (8 / 20) 2 0,5 1 2 4 0,5 1 2 0 0	 μ s voltage (current) Ω kV kV Ω μ F	 IEC 61000-4-5 	 See 7.2.7 	 B
^a These coupling values are for screened communication ports. For unscreened communication ports use coupling values given in Table 6, item 6.4.						

6.4 Immunity input and output ports

Table 6 – Immunity tests – Input and output ports

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
6.1	Conducted disturbance induced by radio-frequency fields					
	Frequency sweep					
	Test frequency range	0,15 to 80	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Sweep rate	≤ 1	% (of preceding frequency value)	IEC 61000-4-6	See 7.2.8	A
	Spot frequencies					
	Test spot frequencies	27 $\pm$ 0,5 % 68 $\pm$ 0,5 %	MHz MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
6.2	Fast transients					
	Rise time t_r / duration time t_d	5 / 50	ns			
	Test level			IEC 61000-4-4	See 7.2.5	B
	Zone A	4	kV peak voltage			
	Zone B	2	kV peak voltage			
6.3	Slow damped oscillatory wave					
	Voltage oscillation frequency	1	MHz			
	Test level					
	Differential mode ^c	1	kV peak voltage	IEC 61000-4-18	See 7.2.6	B
	Common mode	2,5	kV peak voltage			
	Voltage rise time	75	ns			
	Repetition frequency	400	Hz			
	Output impedance	200	Ω			

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
6.4	Surge Front time / time to half value Source impedance Test level Line-to-line Zone A Zone B Coupling resistor Coupling capacitor Line-to-earth Zone A Zone B Coupling resistor ^b Coupling capacitor ^b	 1,2/50 (8/20) 2 0,5 1 2 0,5 1 40 0,5 0,5 1 2 4 0,5 1 2 40 0,5	 μs voltage (current) Ω kV kV Ω μF kV kV Ω μF	 IEC 61000-4-5	 See 7.2.7	 B
6.5	Power frequency ^a Zone A Test voltage differential mode Coupling resistor Coupling capacitor Test voltage common mode Coupling resistor Coupling capacitor Zone B Test voltage differential mode Coupling resistor Coupling capacitor Test voltage common mode Coupling resistor Coupling capacitor	 150 100 0,1 300 220 0,47 100 100 0,047 300 220 0,47	 V r.m.s. Ω μF V r.m.s. Ω μF V r.m.s. Ω μF V r.m.s. Ω μF	 IEC 61000-4-16	 See 7.2.9	 A
^a The power frequency test is only applicable to binary input ports. ^b For screened input/output, e.g. transducer input/output use coupling as per Table 5, item 5.4. ^c In more severe environments a differential test voltage of 2,5 kV may be required for current and voltage transformer inputs.						

6.5 Immunity functional earth port

Table 7 – Immunity tests – Functional earth port

Item	Environmental phenomena	Test specification	Units	Basic standard	Test procedure	Acceptance criteria (see 8.2)
7.1	Conducted disturbance induced by radio-frequency fields					
	Frequency sweep					
	Test frequency range	0,15 to 80	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Sweep rate	≤ 1	% (of frequency)	IEC 61000-4-6	See 7.2.8	A
	Spot frequencies					
	Test spot frequencies	27 $\pm$ 0,5 %	MHz			
		68 $\pm$ 0,5 %	MHz			
	Test level (prior to modulation)	10	V (r.m.s.)			
	Source impedance	150	Ω			
	Amplitude modulated	80	% AM (1 kHz)			
	Duty cycle	100	%			
7.2	Fast transients					
	Rise time t_r / duration time t_d	5 / 50	ns			
	Test level			IEC 61000-4-4	See 7.2.5	B
	Zone A	4	kV peak voltage			
	Zone B	2	kV peak voltage			
	Repetition frequency	5	kHz			

7 Test set-up and procedures

7.1 Emission

7.1.1 General

The tests shall be carried out with the equipment under reference conditions as stated in IEC 60255-1, CISPR 11 and CISPR 22.

The tests shall be carried out with the rated values of auxiliary energizing and input energizing quantities applied to the appropriate circuits of the EUT. The applied values shall be representative of normal in-service quiescent conditions and the EUT shall neither be in a transient operational state nor in an operated state. Half of the binary inputs and half of the output relays shall be energized. Communication modules, if any, shall be activated.

For relays with a wide operating power supply range or rating, the radiated emissions shall be performed on the lowest and the highest nominal voltage of the relay power supply.

The conducted emission shall be performed on all nominal voltages of the relay power supply.

For most of their operational life, measuring relays and protection equipment are in a quiescent state, and the instances when they may be operating or operated are extremely short; in these instances, the emissions from the EUT are not considered to be of significance.

7.1.2 Radiated emission

See Table 8.

Table 8 – Radiated emission test

Test set-up	Test procedure
Test configuration	In accordance with CISPR 11 for radiated emission below 1 GHz and in accordance with CISPR 22 for radiated emission above 1 GHz
Distance and method	See Table 1
Details of mounting	Installed according to manufacturer's specification
Frequency range	See Table 1
Acceptance criteria Class A limits	See Table 1

7.1.3 Conducted emission

See Table 9.

Table 9 – Conducted emission test

Test set-up	Test procedure
Test configuration	In accordance with CISPR 22
Applicable ports	A.C. and d.c. auxiliary power supply port
Details of mounting	Installed according to manufacturer's specification
Frequency range	See Table 2
Acceptance criteria Class A limits	See Table 2

7.2 Immunity

7.2.1 General

The general test set-up, i.e. the test generator, coupling and decoupling devices for each immunity test is specified in the appropriate basic standard for each test.

7.2.2 details the general conditions that shall be complied with when carrying out immunity testing to measuring relays and protection equipment.

7.2.3 to 7.2.13 detail the test procedures and specific applications of the immunity tests that shall be applied to measuring relays and protection equipment.

7.2.2 General test conditions

7.2.2.1 Test reference condition

The test reference conditions shall be as stated in IEC 60255-1 unless the relevant basic standard of the IEC 61000-4 series referred to in this standard, states test reference conditions that have stricter tolerances.

Where there are many input or output ports of identical circuits on a board or module, such as binary inputs or output contacts and there are many such boards in the EUT and the test is applied directly to the circuits, then it shall be sufficient to test only three such circuits on each of the board or module in each slot in order to claim compliance with that particular test. This only applies to immunity tests that are applied to ports of the EUT rather than the enclosure as a whole.

7.2.2.2 Configuration

The EUT shall be configured and set-up in accordance with the manufacturer's recommendations in such a way that it is representative of the way that the EUT will be used in normal service.

Input and output ports shall be connected and energized as appropriate in line with normal service conditions and installation practice.

The EUT communication port(s) shall be connected and configured in accordance with the manufacturer's recommendations.

NOTE Some guidance on testing is given in Annex A of IEC 60255-1:2009.

7.2.2.3 Cable specification

The cable type used for each input and output port shall be as recommended by the manufacturer. Where there is a restriction on the maximum cable length used on an input or output port this shall be stated by the manufacturer. The maximum cable length used shall be as recommended for each test according to the basic standard.

7.2.2.4 Support and monitoring equipment

Auxiliary support and monitoring equipment shall be connected to the EUT to provide signals to exercise all the functions of the EUT and to monitor the EUT during the application of the test. The support and monitoring equipment shall be chosen so that it does not have any adverse influence on the test and is not affected by the interference applied during the immunity tests. Where the application of the test does affect the auxiliary support and monitoring equipment suitable filtering and de-coupling networks shall be used that do not affect the applied test. The values and examples of filtering and de-coupling networks are contained within the basic standards.

7.2.2.5 Settings

Protection settings and functional settings shall be applied to the EUT with reference to IEC 60255-1. The delay settings shall be set to their minimum values. The immunity tests shall be carried out with the EUT in the quiescent state and where indicated in the test procedure in the operated state also. For current transformer inputs the most sensitive rating shall be used.

7.2.2.6 Functional tests

Functional tests to verify the correct operation of the EUT within its accuracy rating shall be carried out before and after the application of the immunity tests. Some individual tests shall also require functional tests to be carried out during the application of the immunity test.

Whenever this is required it will be stated in the test procedure for the appropriate test. In case of multifunctional relays the manufacturer should at least dedicate one protection function, normally the main function, for checking the adequate operation during EMC immunity tests.

7.2.2.7 Test generator verification

The output of the test generator shall be verified as stated in the basic standard.

7.2.3 Electrostatic discharge

See Table 10.

Table 10 – Electrostatic discharge immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2
Initial measurement	The tests shall be carried out in accordance to 7.2.2
Details of mounting/support	The EUT is placed in its case or housing in as close to installed conditions as possible
Discharge method	Tests shall be applied as follows: – the contact discharge method is the preferred method; – the air discharge method shall only be used when the accessible surfaces of the EUT are non-conducting; – the direct and indirect application test method shall be used.
Application of the discharges	See Annex C
Test voltage	See Table 3, item 3.2
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the electrostatic discharges. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The values of input energizing quantities shall be within twice the assigned error below of the transitional state, see IEC 60255-1:2009, Annex A. Since the coincidence of electrostatic discharges and a primary fault is considered to be unlikely, the effect of the electrostatic discharges on the EUT in its transitional or operate condition is not considered. The points selected for the application of the test shall be those which are accessible to the operator under normal service conditions, including communication ports and those points for setting adjustments that can only be accessed by removing the relay cover. Setting adjustments which necessitate any action other than removal of the cover, such as removal of a module, are not included. The application of discharge to any point of the equipment which is accessible only for repair and maintenance purposes is outside the scope of this standard. In the selection of test points, attention shall be given to the following: - knobs, push-buttons, switches, communication interface, etc., accessible under normal service; - points on covers of insulating material where conducting parts are close to the inside of the cover; - points on conducting parts not belonging to, but placed in the vicinity of, the EUT, when this has an insulating cover. To achieve reproducible results, it is recommended that the selected test points are specified by the manufacturer (see Annex C). Testing shall be satisfied at all levels given in Table 3, item 3.2.
Acceptance criteria	See Table 3, item 3.2.

7.2.4 Radiated interference

7.2.4.1 Frequency sweep

See Table 11.

Table 11 – Radiated interference immunity test (frequency sweep)

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	The EUT is placed in the calibrated test field.
Frequency range to be swept	See Table 3, item 3.1.
Modulation	See Table 3, item 3.1.
Dwell time	The dwell time at each frequency shall be 0,5 s. In those cases where the EUT operate time is greater than 0,5 s, the dwell time shall be increased until operation of the EUT is possible.
Test field strength	See Table 3, item 3.1
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the electromagnetic field. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its transitional or operate state is not considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state, see IEC 60255-1:2009, Annex A.
Acceptance criteria	See Table 3, item 3.1.

7.2.4.2 Spot frequencies

See Table 12.

Table 12 – Radiated interference immunity test (spot frequencies)

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	The EUT is placed in the calibrated test field.
Spot frequencies	See Table 3, item 3.1.
Modulation	See Table 3, item 3.1.
Duty cycle	See Table 3, item 3.1.
Test duration	≥ 10 s
Dwell time	The dwell time at each spot frequency shall be sufficient for the EUT to change from the quiescent state to the operate state.
Test field strength	See Table 3, item 3.1.
Measurement and verification during the test	The EUT is capable of correct operation and reset in the presence of an electromagnetic field from a radiation source at the given spot frequencies. During each spot frequency test, the input energizing quantities shall be adjusted to cause the EUT to change from the normal energized state to the operated state, and held until the EUT operates correctly. The input energizing quantities shall then be re-adjusted to cause the EUT to reset.
Acceptance criteria	See Table 3, item 3.1.

7.2.5 Electrical fast transient

See Table 13.

Table 13 – Electrical fast transient immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	All auxiliary equipment used to provide the EUT with signals for normal operation, and to verify the correct operation of the EUT, shall be decoupled. For the functional earth port, the fast transient/burst test is not applicable to this port if interfacing with cables whose total length according to the manufacturer's functional specification is always less than 3 m. The fast transient/burst test is not applicable to the communications port if interfacing with cables which are, in normal use, permanently connected, and whose total length according to the manufacturer's functional specification is always less than 3 m. Where the EUT is mounted in a cubicle, the tests may be conducted in the cubicle. No test shall be performed on interconnecting cables within the same cubicle.
Application ports/methods	
Auxiliary power supply port	CDN
AC current and voltage ports	CDN
Functional earth port	Capacitive coupling clamp
Binary inputs	Capacitive coupling clamp
Output contacts	Capacitive coupling clamp
Communication ports	Capacitive coupling clamp
Test voltage	See Table 4, item 4.2, Table 5, item 5.2, Table 6, item 6.2, Table 7, item 7.2.
Repetition frequency	See Table 4, item 4.2, Table 5, item 5.2, Table 6, item 6.2, Table 7, item 7.2.
Test wave form characteristic	See Table 4, item 4.2, Table 5, item 5.2, Table 6, item 6.2, Table 7, item 7.2.
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the disturbance. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its quiescent and operate state is considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state, see IEC 60255-1:2009, Annex A. The test voltage shall be applied in common mode to one port at a time for at least 1 min for each polarity.
Acceptance criteria	See Table 4, item 4.2, Table 5, item 5.2, Table 6, item 6.2, Table 7, item 7.2

7.2.6 Slow damped oscillatory wave

See Table 14.

Table 14 – Slow damped oscillatory wave immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	All auxiliary equipment used to provide the EUT with signals for normal operation, and to verify the correct operation of the EUT, shall be decoupled. Where the EUT is mounted in a cubicle, the tests may be conducted with the EUT in the cubicle.
Waveform	The damped oscillating wave shall be the envelope of the 5 th peak which shall be > 50% of the initial peak value and the 10 th peak which shall be < 50% of the initial peak value.
Voltage rise time	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
Voltage oscillation frequency	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
Source impedance	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
Frequency repetition	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
Test duration	2 s minimum
Length to connection	2 m maximum
Test level	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
Application ports/methods	
Auxiliary power supply port ^a	CM/DM with CDN
Input and output ^a	CM/DM with CDN
Communication ports ^b	CM
Measurement and verification during the test ^c	The EUT will have normal performance within the specification limits when energized and subjected to the disturbance. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its quiescent and operate state is considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state, see IEC 60255-1:2009, Annex A.
Acceptance criteria	See Table 4, item 4.3, Table 5, item 5.3 and Table 6, item 6.3.
^a Common mode test shall be performed between each independent port and earth and between each independent port and all other independent ports coupled to earth. ^b The damped oscillatory wave test is not applicable to the communication port if interfacing with cables which are not permanently connected, or whose total length according to the manufacturer's functional specification is always less than 3 m. ^c Where the operating time of the EUT is longer than 2 s, the test voltage shall be applied for a period longer than the actual operating time of the EUT. The minimum time interval between two successive tests shall be 1 s.	

7.2.7 Surge

See Table 15.

Table 15 – Surge immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2
Initial measurement	The tests shall be carried out in accordance to 7.2.2
Details of mounting/support	The EUT shall be configured in accordance with 7.2.2 The EUT shall be tested individually on a bench or table top. Where the EUT is mounted in a cubicle and is only used with other devices then the tests may be conducted with the EUT in the cubicle. Any requirements for insulating supports for the EUT as stated in the basic standard shall be followed. The support equipment shall be configured in accordance with 7.2.2, any requirements for insulating supports for the support equipment and interconnecting cables as stated in the basic standard shall be followed. The connections between the EUT and the test generator shall be less than 2 m, and except in the case of the testing of the communications port, connections between the EUT and the coupling/decoupling networks shall also be less than 2 m. The maximum cable length of screened interfaces shall be not more than 20 m.
Application ports and values of source impedance and coupling capacitance ^{a b c d}	
Auxiliary power supply port	See Table 4, item 4.4.
AC current and voltage ports	See Table 6, item 6.4.
Binary inputs	See Table 6, item 6.4.
Output contacts	See Table 6, item 6.4.
Communication ports ^{a, e}	See Table 5, item 5.4.
Test waveform characteristic	See Table 4, item 4.4, Table 5, item 5.4, Table 6, item 6.4.
Test voltages	See Table 4, item 4.4, Table 5, item 5.4, Table 6, item 6.4.
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the surge immunity test. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its transitional or operate state is not considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state, see IEC 60255-1:2009, Annex A. The number of tests shall be at least five positive and five negative surges. The repetition rate shall be a maximum of 1 surge/min. Because of the possibility that the EUT has non-linear current-voltage characteristics, all lower test voltages up to and including the maximum test voltage selected shall be satisfied. The surges shall be applied line to earth and line to line where appropriate. When testing line to earth the test voltage shall be applied successively between each of the lines and earth.
Acceptance criteria	See Table 4, item 4.4, Table 5, item 5.4, Table 6, item 6.4
^a Unless otherwise stated, no test is advised for ports interfacing with cables whose total length according to the manufacturer's functional specification is always less than 10 m.	

- ^b No line to line test is advised on input and output ports which according to the manufacturer's functional specification are always interfacing with twisted pair screened cables.
- ^c The application of the surge is to simulate a lightning strike and hence is a short duration high energy pulse. Low impedance high sensitivity circuits may produce a start signal when such a pulse is applied. If this is the case the relay (EUT) has interpreted the pulse correctly as per their design. In that case the manufacturer shall state any time delays or setting limitations which should be applied to ensure immunity to the surge pulse.
- ^d Where the coupling capacitor has an effect on the operation of the circuit under test, alternative methods of coupling networks are allowed as described in the basic standard, examples of such devices are a gas discharge tube or a varistor. Where such an alternative coupling network is used it should be stated in the test report.
- ^e For unscreened communication ports use coupling as specified in Table 6, item 6.4.

7.2.8 Conducted interference

7.2.8.1 Frequency sweep

See Table 16.

Table 16 – Conducted interference immunity test (frequency sweep)

Test set-up	Test procedure
EUT configuration	See 7.2.2.2
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	All auxiliary equipment used to provide the EUT with signals for normal operation and to verify the correct operation of the EUT, shall be decoupled. Where the EUT is mounted in a cubicle, the test may be conducted in the cubicle. No test shall be performed on interconnecting cables within the same cubicle.
Application ports/methods	For coupling method on different application ports, see IEC 61000-4-6.
Test level	See Table 4, item 4.1, Table 5, item 5.1, Table 6, item 6.1, Table 7, item 7.1.
Frequency range to be swept	See Table 4, item 4.1, Table 5, item 5.1, Table 6, item 6.1, Table 7, item 7.1.
Amplitude modulation	See Table 4, item 4.1, Table 5, item 5.1, Table 6, item 6.1, Table 7, item 7.1.
Dwell time	The dwell time at each frequency shall be not less than 0,5 s. In those cases where the EUT operate time is greater than 0,5 s, the dwell time shall be increased until operation of the EUT is possible.
Test level	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the conducted disturbance. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its transitional or operate state is not considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state.
Acceptance criteria	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.

7.2.8.2 Spot frequencies

See Table 17.

Table 17 – Conducted interference immunity test (spot frequencies)

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	All auxiliary equipment used to provide the EUT with signals for normal operation and to verify the correct operation of the EUT, shall be decoupled. Where the EUT is mounted in a cubicle, the test may be conducted in the cubicle. No test shall be performed on interconnecting cables within the same cubicle.
Application ports/methods	For coupling method on different application ports see IEC 61000-4-6.
Spot frequencies	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.
Amplitude modulation	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.
Duty cycle	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.
Test duration	≥ 10 s
Dwell time	The dwell time at each spot frequency shall be sufficient for the EUT to change from the quiescent state to the operate state.
Test level	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.
Measurement and verification during the test	The EUT is capable of correct operation and reset in the presence of the conducted disturbance at the given spot frequencies. During each spot frequency test, the input energizing quantities shall be adjusted to cause the EUT to change from the normal energized state to the operated state, and held until the EUT operates correctly. The input energizing quantities shall then be readjusted to cause the EUT to reset.
Acceptance criteria	See Table 4, item 4.1; Table 5, item 5.1; Table 6, item 6.1 and Table 7, item 7.1.

7.2.9 Power frequency immunity on d.c. binary inputs

See Table 18.

Table 18 – Power frequency immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	The EUT is placed in normal operating condition.
Application ports and values of source impedance and coupling capacitance ^{a b}	See Table 6, item 6.5.
Test frequency	Rated frequency, e.g. 16,7 Hz, 50 Hz or 60 Hz.
Test voltage	See Table 6, item 6.5.
Measurement and verification during the test	The tests shall be carried out with auxiliary energizing quantities equal to rated values applied to the auxiliary d.c. power supply port. The test voltage shall only be applied with the d.c. binary input not energized to verify that it does operate correctly. If the d.c. binary input has software or hardware controllable delay, then the test voltage should be applied first with the delay set to its minimum value. If this results in failure, then the delay value should be increased and the test voltage re-applied until the test is passed; the value of this final d.c. binary input delay shall be recorded in the test report.
Acceptance criteria	See Table 6, item 6.5.
NOTE 1 Annex A contains technical information for the power frequency immunity tests.	
NOTE 2 Annex B contains background information for the power frequency immunity tests.	
^a The differential mode test is not required on d.c. binary input ports which, according to the manufacturer's functional specification, always have an interface via multi-core screened or twisted pair (screened or unscreened) cables. ^b Unless otherwise stated, no test is required for d.c. binary input ports interfacing with cables whose total length according to the manufacturer's functional specification is always less than 10 m.	

7.2.10 Power frequency magnetic field

See Table 19.

Table 19 – Power frequency magnetic field immunity test

Test set-up	Test procedure
EUT configuration	See 7.2.2.2.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Details of mounting/support	The EUT is placed in the magnetic field of a 1 m × 1 m induction coil.
Test frequency	Rated power frequency of the EUT, which is used where the EUT is intended to be installed, (e.g. 50 Hz or 60 Hz).
Test duration	Continuous: ≥ 60 s Short time: 1 s to 3 s
Test field strength	See Table 3, item 3.3.
Measurement and verification during the test	The EUT will have normal performance within the specification limits when energized and subjected to the magnetic field. The tests shall be carried out with auxiliary energizing quantities applied to the appropriate circuits, using input energizing quantities equal to rated values. The effect of this disturbance on the relay in its quiescent and operate state is considered. The values of the input energizing quantities shall be within twice the assigned error of the transitional state, see IEC 60255-1:2009, Annex A.
Acceptance criteria	See Table 3, item 3.3.

7.2.11 Voltage dips and voltage interruptions on power supply voltage (a.c. or d.c.)

See Table 20.

Table 20 – Voltage dips and voltage interruptions test

Test set-up	Test procedure
EUT configuration	Accessories shall be connected according to manufacturer's specification. EUT shall be in a quiescent state. Half of the binary inputs and half of the output relays shall be energized. Communication modules, if any, shall be activated.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Complementary test procedure	
Selection of application points	The tests shall be applied to power supply terminals.
Test levels	See Table 4, item 4.5 and item 4.6.
Number of trials	3
Time interval between trials	≥10 s
Measurement and verification during the test	The voltage in this standard uses the rated voltage for the EUT as a basis for voltage test level specification. Where the EUT has a rated voltage range, the test procedure shall be applied at the lowest voltage declared in the voltage range. Example: EUT with a rated voltage range of 100 V to 200 V ± 20 % should be tested at 80 V. For EUT intended to be powered by d.c. power supply, only the corresponding tests shall be done. For EUT intended to be used with a.c. power supply, only the corresponding tests shall be done. For EUT intended to be used either with a.c. power supply or with d.c. power supply both tests shall be done.
Acceptance criteria	See Table 4, item 4.5 and item 4.6.

7.2.12 Voltage ripple on d.c. power supply voltage

See Table 21.

Table 21 – Voltage ripple test

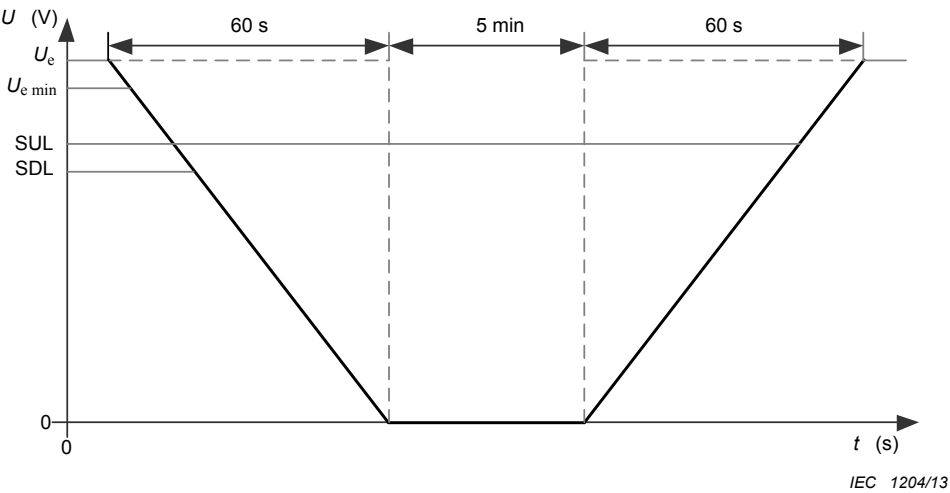
Test set-up	Test procedure
EUT configuration	Accessories shall be connected according to manufacturer's specification. EUT shall be in a quiescent state. Half of the binary inputs and half of the output relays shall be energized. Communication modules, if any, shall be activated.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Complementary test procedure	
Selection of application points	The tests shall be applied to power supply terminals.
Test level	See Table 4, item 4.7.
Test duration	1 min
Measurement and verification during the test	The voltage in this standard uses the rated voltage for the EUT as a basis for voltage test level specification. Where the EUT has a rated voltage range, the test procedure shall be applied for both the lowest and highest voltage declared in the voltage range. Example: EUT with a rated voltage range of 100 V to 200 V $\pm$ 20 % should be tested at 80 V and 240 V.
Acceptance criteria	See Table 4, item 4.7.

7.2.13 Gradual shut down / start-up tests

See Table 22.

Table 22 – Gradual shutdown and start-up test

Test set-up	Test procedure
EUT configuration	Accessories shall be connected according to manufacturer's specification. EUT shall be in a quiescent state. Half of the binary inputs and half of the output relays shall be energized. Communication modules, if any, shall be activated.
Initial measurement	The tests shall be carried out in accordance to 7.2.2.
Complementary test procedure	
Selection of application points	The tests shall be applied to power supply terminals.
Test level	See Table 4, item 4.8
Measurement and verification during the test	The voltage in this standard uses the rated voltage for the EUT as a basis for voltage test level specification. Where the EUT has a rated voltage range, the test procedure shall be applied at the lowest voltage declared in the voltage range. Example: EUT with a rated voltage range of 100 V to 200 V ± 20 % should be tested at 80 V (see Figure 2).
Acceptance criteria	See Table 4, item 4.8.



Key

- U_e rated auxiliary power supply voltage
- $U_{e\ min}$ lower limit of U_e
- SDL shut-down limit
- SUL start-up limit

Figure 2 – Gradual shut down/start-up test

8 Criteria for acceptance

8.1 Emission

The EUT shall satisfy the requirements of this specification if conducted and radiated emissions during the tests do not exceed the limits given in Table 1 and Table 2.

The results of an evaluation of the emissions of EUTs having at least one of each type of identical module can be applied to configurations having more than one of each type of those modules. This is permissible because it has been found that, in practice, emissions from identical modules are not additive. This non-additive principle can be applied to protection equipment made up of a number of identical measuring relays.

8.2 Immunity

See Table 23.

Table 23 – Acceptance criteria for immunity tests

Criteria	Function	Conditions for acceptance
A	Protection	Normal performance within the specification limits, during and after the test.
	Command and control	Normal performance within the specification limits, during and after the test.
	Measurement	No degradation during test.
	Integral human-machine interface and visual alarms	No degradation or no loss of function during test. No loss of stored data.
	Data communication ^b	Possible bit error rate increase but no loss of transmitted data.
	Binary input, binary output and output contacts	No unwanted change of status is allowed during the test ^a .
B	Protection	Normal performance within the specification limits, during and after the test.
	Command and control	Normal performance within the specification limits, during and after the test.
	Measurement	Temporary degradation during test, with self-recovery at the end of the test. No loss of stored data.
	Integral human-machine interface and visual alarms	Temporary degradation or loss of function during test, with self-recovery at the end of the test. No loss of stored data.
	Data communication ^b	Possible bit error rate increase but no loss of transmitted data.
	Binary input, binary output and output contacts	No unwanted change of status is allowed during the test ^a .
C	Protection	Temporary loss of function provided the function is self-recoverable. No unwanted operation shall be observed.
	Command and control	Temporary loss of function provided the function is self-recoverable. No unwanted operation shall be observed.
	Measurement	Temporary loss of function provided the function is self-recoverable.
	Integral human-machine interface and visual alarms	Temporary loss of function provided the function is self-recoverable.
	Data communication ^b	Temporary loss of function, provided the function is self-recoverable Possible loss of transmitted data.
	Binary input, binary output and output contacts	No unwanted change of status is allowed during the test ^a .
If the manufacturer is using a specification for degradation during or after the testing required by this standard, the specification shall be provided in the product documentation available to the user.		
^a For binary inputs the manufacturer shall state the minimum filtering value for which the test was successful.		
^b Excluding communication ports for protection or control functionality. For those acceptance criteria see protection or command and control.		

9 Test report

A test report giving the test procedures and results shall always be produced.

The test report shall include at least the following basic information:

- a) a title (e.g. “test report”);
- b) the name and address of the laboratory, and the location where the tests were carried out, if different from the address of the laboratory;
- c) a unique identification of the test report (such as the serial number), and on each page an identification that helps to recognize the page as part of the test report and a clear identification of the end of the test report;
- d) the name and address of the client;
- e) a description of the condition of the EUT and its unambiguous identification;
- f) the date(s) of performance of the test;
- g) the test results with, where appropriate, the units of measurement;
- h) the name(s), function(s), and signature(s) or equivalent identification of person(s) authorizing the test report;
- i) where relevant, a statement to the effect that the results relate only to the EUT.

In addition to the above basic information, test reports shall include the following information:

- j) the test conditions;
- k) the test methods;
- l) the measuring equipment used;
- m) the test conclusion(pass/fail);
- n) where appropriate and needed, opinions and interpretations.

Annex A (normative)

Power frequency immunity tests on binary inputs

A.1 General

The power frequency immunity tests are based on the concepts described in IEC 61000-4-16, referring to that publication where applicable. The objective of the tests is to confirm that the equipment under test (EUT) will operate correctly when energized and subjected to short duration, conducted, common and differential mode power frequency disturbances applied to d.c. binary inputs, at the rated frequency of the EUT, for example 16,7 Hz, 50 Hz or 60 Hz.

The testing of pilot wire schemes between substations is not covered by these tests.

A.2 Test classes

Class A test levels are applicable to substations with high earth fault currents, and where the standard wiring practices allows the d.c. binary inputs to be wired to primary plant auxiliary contacts via 'open' loops. An open loop occurs where the go and return leads are allowed in different multi-core cables, and therefore face the risk of following substantially different paths. This produces a large potential area of magnetic flux linkage with the primary earth fault current, which causes high levels of power frequency interference.

Class B test levels are applicable to either:

- substations with low earth fault currents, for example substations which use isolated or Petersen coil earthing; or
- where wiring practice guarantees that the d.c. binary inputs are not wired in 'open' loops; open loops are avoided by using the go and return wires in the same multi-core cable; this ensures that the go and return routes are essentially identical, and that the area of magnetic flux linkage with the primary earth fault current is small, thus minimizing the levels of power frequency interference.

A.3 Test equipment

A.3.1 General

Where the test frequency is not that of the available mains distribution network, an alternative test generator will have to be used, for example as described in 6.1.3 of IEC 61000-4-16:1998, Amendment 2:2009.

A.3.2 Test generator

The test generator typically consists of a variable transformer connected to the mains distribution network and an isolation transformer. The generator shall have the following characteristics:

Waveform:	Sinusoidal, total harmonic distortion less than 10 %.
Open circuit output voltage range:	100 V to 300 V r.m.s. (± 10 %).
Impedance:	Less than 150 Ω .

Frequency:	Selected rated frequency ($\pm 0,5$ Hz).
On/off output voltage switching:	Synchronized at zero crossing ($0^\circ \pm 10^\circ$) or increased from zero / decreased to zero (see A.4.3).

A.3.3 Verification of the test generator

In order to ensure that the results when using different test generators can be meaningfully compared, the following characteristics of the generator shall be calibrated or verified:

- output voltage waveform;
- voltage generator impedance;
- frequency accuracy;
- open circuit output voltage accuracy.

The verifications shall be carried out with a voltage probe and oscilloscope or other equivalent measuring instruments with a minimum bandwidth of 1 MHz. The accuracy of these instruments shall be better than ± 5 %.

A.3.4 Coupling networks

The coupling networks enable the test voltage to be applied in both common and differential mode (see Figures A.1, A.2 and A.3 for typical test set-ups).

The networks consist of a resistor and capacitor in series. The values for these components for the tests are shown in Table 6, item 6.5, and the values for each pair of capacitors and resistors shall be matched with a tolerance of 1 %.

A.4 Test set-up

A.4.1 General

Figures A.1 and A.2 show typical test set-ups for differential mode tests, and Figure A.3 shows a typical test set-up for common mode tests. Connections between the EUT and the coupling network shall be less than 2 m.

A.4.2 Earthing connections

The EUT, the auxiliary equipment and the test equipment shall comply with their safety earthing requirements at all times. Additionally, the EUT shall be connected to the earthing system in accordance with the manufacturer's specifications.

A.4.3 Auxiliary equipment

All auxiliary equipment, used to provide the EUT with signals for normal operation, and to verify the EUT's correct operation, shall be decoupled so that the test voltage does not affect the auxiliary equipment.

Auxiliary equipment required for the EUT to operate as defined by the specifications (e.g. communication equipment, modem, printer, etc.), as well as auxiliary equipment necessary for ensuring any data transfer and verification of the functions, shall be connected to the EUT. However, as far as possible the number of cables to be monitored will be limited by taking into consideration only the representative functions.

The test generator shall be connected to the d.c. binary input port of the EUT. When this port consists of multiple identical circuits, only a representative number of these circuits, as

defined by the manufacturer, need to be tested in order to verify the correct operation of the EUT.

The test voltage shall be applied for at least 10 s so that the EUT's operating performance can be verified. The test voltage shall be applied as shown in Figures A.1, A.2 and A.3.

When a test generator with zero crossing synchronization is not available, in order to avoid unwanted transients at turn on and turn off, the test voltage may be increased from zero to the required level at the start of the test, and decreased to zero at the end of the test. The duration of these start and end phases shall not be included as part of the test time, and they shall each have duration of less than 20 % of the time of application of the required test voltage.

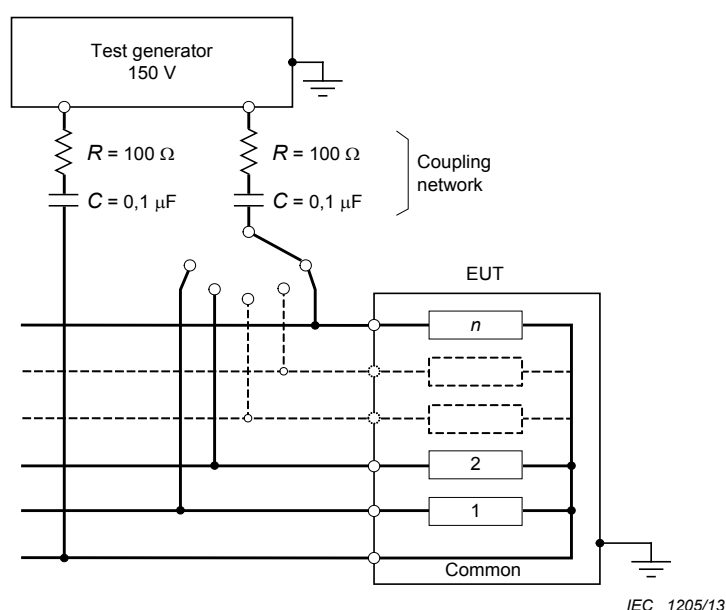


Figure A.1 – Example of Class A differential mode tests

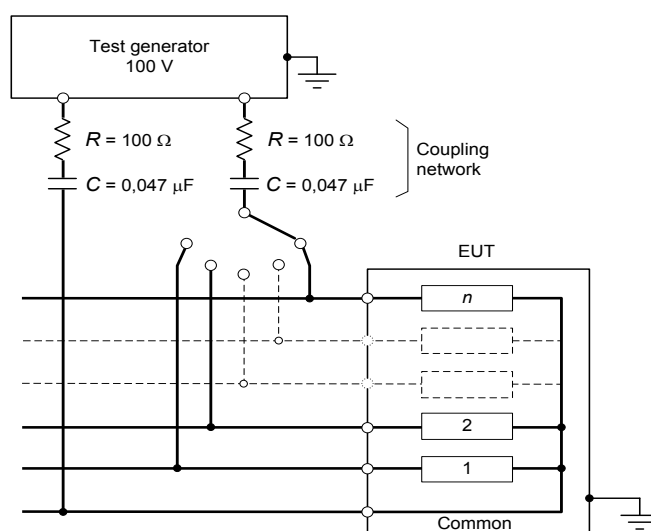


Figure A.2 – Example of Class B differential mode tests

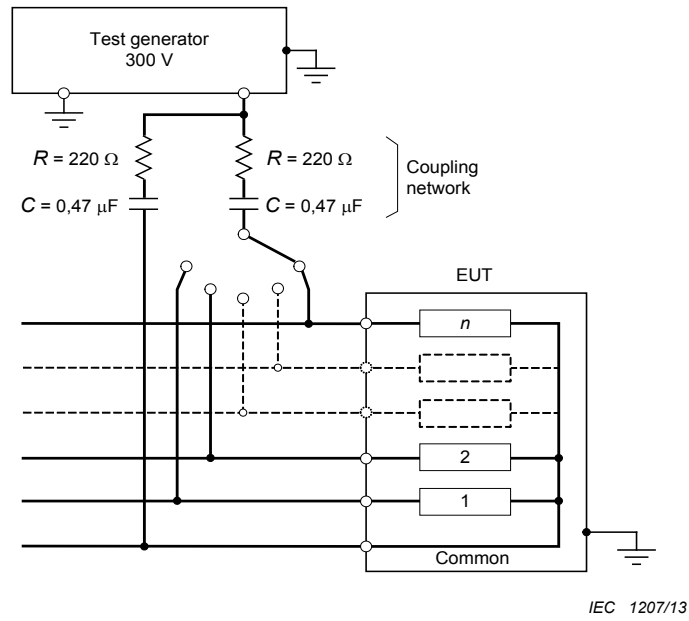


Figure A.3 – Example of common mode tests

Annex B (informative)

Background information for power frequency tests

Conducted interference voltages are generated by different sources of interference, and can be transferred by inductive or capacitive coupling to the supply cables, signal cables and earthing of measuring relays and protection equipment.

The electronic environment in which the equipment is used is also related to the sources of interference which may be present in different kinds of installations, for example substations, and is also related to the coupling which is given by the normal installation of the equipment, i.e. power supply, location, type of cables, earthing, screening, filtering, etc.

In the case of an earth fault in a substation, a high current passes through the earthing system, and thus the electrical potential of different parts of the substation will be raised with respect to each other and to earth. This means that the cables used for signaling between equipment are subjected, in the case of balanced systems, to common mode voltages at mains frequency. Additionally, in unbalanced circuits, a differential mode voltage is generated, the magnitude of which depends on the unbalance in the input circuit within the equipment and the physical arrangement of the signal cabling.

Similar interference voltages of mains frequency can also be induced in signal cables without any earth fault current, for example if a power cable and a signal cable are laid parallel to and close to each other.

It is recognized that these types of interference will to some degree appear on all copper circuits within a substation, and it is these interference voltages which this power frequency test attempts to simulate.

It should be noted that although capacitive coupling is used to apply the test voltage to the signal cabling, this test remains valid for simulating both capacitive and inductively coupled interference, as both these types of interference result in an induced voltage on the signal cables.

Annex C (normative)

Application of discharges for electrostatic discharge test

The contact discharge method of testing (the preferred method) shall be applied to conductive surfaces of the EUT.

The air discharge method shall only be used when the accessible surfaces of the EUT are non-conducting.

The test voltage shall be increased from the minimum to the selected test level, in order to determine any threshold of failure.

It is recommended that the discharge generator includes a monitor device for indication of the actual application of the discharge at the corresponding level.

The test shall be performed with single discharges. It shall be repeated on each selected test point at least 10 times with positive polarity and 10 times with negative polarity of test voltage.

The recommended time interval between successive single discharges is 1 s, although longer intervals may be necessary to determine whether a failure has occurred.

The points to which the discharges shall be applied may be selected by means of an exploration carried out at a repetition rate of 20 discharges per second, or more.

The discharge generator shall be held perpendicular to the surface to which the discharge is applied. This improves repeatability of the test results.

The discharge return cable of the generator shall be kept at a distance of at least 0,2 m from the EUT and metal surfaces, other than the ground plane, whilst the discharge is being applied.

For contact discharge tests, the tip of the discharge electrode shall touch the EUT before the discharge switch is operated.

In the case of painted surfaces covering a conducting substrate at the intended point of test, the following procedure shall be adopted:

- where the coating is not declared to be an insulating coating by the equipment manufacturer, then the pointed tip of the generator shall penetrate the coating so as to make contact with the conducting substrate;
- where the coating is declared as insulating with a declared withstand voltage exceeding the appropriate test voltage level for air discharge tests, the contact discharge test shall not be applied to such surfaces.

For air discharge tests, the discharge tip of the discharge electrode shall approach and touch the EUT as fast as possible without causing mechanical damage. After each discharge, the discharge electrode shall be removed from the EUT, and the generator shall be retriggered in readiness for the next single discharge. This procedure shall be repeated until the discharges are completed.

For test method of ungrounded equipment, see IEC 61000-4-2.

Bibliography

IEC 61000-4 (all parts), *Electromagnetic compatibility (EMC) – Part 4: Basic standards*

IEC/TS 61000-6-5, *Electromagnetic compatibility (EMC) – Part 6-5: Generic standards – Immunity for power station and substation environments*

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